

Problem F. Refrigerator

Input file: **standard input**
Output file: **standard output**
Time limit: **2 seconds**
Memory limit: **256 megabytes**

- *Major, to use the refrigerator “Melania”, enter the password.*
- *What else is the password?*
- *Now I’ll tell you everything...*

You are given two numbers n and q and an array a of n positive integers. To open the refrigerator you need to correctly answer q queries:

1. Add query characterized by numbers l , r and x : add x to all numbers with indices from l to r .
2. Query the sum: given numbers l and r find the value $\sum_{i=1}^{r-l+1} fib_i \cdot a_{l+i-1}$ modulo $10^9 + 7$.

Please note that advanced research by our scientists has shown that the *fib* sequence is optimally specified as follows:

$$fib_i = \begin{cases} 1, & i \leq 2 \\ fib_{i-1} + fib_{i-2}, & i > 2 \end{cases}$$

And we ask you to stick to this method.

Input

The first line contains the numbers n and q — the number of elements in the array a and the number of queries, respectively.

The second line contains n integers — elements of array a .

The following q lines describe queries:

For the first type query: ‘+’ character and three numbers l , r , x ($1 \leq l \leq r \leq n$).

For the second type of query: the ‘?’ character and two numbers l and r ($1 \leq l \leq r \leq n$).

$$1 \leq n \leq 2 \cdot 10^5$$

$$1 \leq q \leq 2 \cdot 10^5$$

$$1 \leq a_i \leq 10^9$$

$$1 \leq x \leq 10^9$$

Output

For each ‘?’ query display one number - answer modulo $10^9 + 7$.

Examples

standard input	standard output
5 6 1 1 1 1 1 ? 1 5 + 1 5 1 ? 1 5 + 1 3 2 + 3 5 3 ? 1 5	12 24 62
5 3 8 7 1 3 4 + 1 4 10 ? 1 5 ? 3 4	116 24